

SolutionArchitecture

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TeamID	XXXXXXX
ProjectName	EduGenie – Google Gemini Powered Learning Assistant
MaximumMarks	5Marks

SolutionArchitectureDiagram

The EduGenie – Google Gemini Powered Learning Assistant follows a layered architecture consisting of the Presentation Layer, Application Layer, AI Processing Layer, and Data Storage Layer. Students interact with the web interface to submit learning requests. The FastAPI server processes these requests and communicates with Google Gemini 1.5 Pro and LaMini-Flan-T5 models to generate educational responses. The generated results are stored and displayed through the application interface.

Presentation/ClientLayer

HTML5 • CSS3 • JavaScript •
Responsive Web UI



APIGateway/Server

Python • FastAPI • Uvicorn



Authentication

User Login •
Session Management

Core Logic

Concept
Explanation •
Question Answering
• Quiz Generation
Text Summarization
• Personalized

External APIs

Google Gemini 1.5
Pro • LaMini-Flan-T5



Data/StorageLayer

JSON Files • User History • Learning Records • Local Storage

ComponentDescriptionTable

Component Name	Description/RoleinArchitecture	TechnologiesUsed
Presentation Layer	Provides the user interface for students and teachers th explanation, Q&A, quiz generation, summarization, and	HTML5, CSS3, JavaScript
APIGateway	Receives requests from the frontend, validat and returns generated responses.	s inputPython, FastAPI, rou es requests to services, Uvicorn
CoreLogic Service	Processes user requests and manages concept expl summarization, and personalize	Python, learning path modules.FastAPI
Database/ Storage	Stores user information, learning history, ge recommendations, and application data.	erated quizzes, summaries, JSON Files, local Storage